

K-Means

Objetivo:
$$\min_{r_{nk} \in \{0,1\}, \sum_k r_{nk} = 1} \sum_n \sum_k r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$
(asignación *hard*)

Inicialización $\boldsymbol{\mu}_k$ (random o K-Means++)
 $\approx$ **E-Step:** $z_n^* \leftarrow \arg \min_k \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$
 $\approx$ **M-Step:** $\boldsymbol{\mu}_k \leftarrow \frac{1}{N_k} \sum_{n: z_n^* = k} \mathbf{x}_n$

K-Means++: $p(\boldsymbol{\mu}_{t+1} = \mathbf{x}_n) = \frac{d_t(\mathbf{x}_n)}{\sum_{n'} d_t(\mathbf{x}_{n'})}$
 $d_t(\mathbf{x}) = \min_{k=1}^K \|\mathbf{x} - \boldsymbol{\mu}_k\|_2^2$
Límite de GMM: $\boldsymbol{\Sigma}_k = \varepsilon \mathbf{I}, \varepsilon \rightarrow 0 \Rightarrow$ asignación hard (gaussianas infinitamente concentradas).

Gaussian Mixture Model (GMM)

$p(\mathbf{x}|\mathbf{w}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k); \sum_k \pi_k = 1$
Latente: $z \sim \text{Cat}(\boldsymbol{\pi}); \mathbf{x}|z = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$
 $\ln p(\mathbf{X}|\mathbf{w}) = \sum_n \ln \sum_k \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$

E-step (responsabilidades, soft clust.):
$$r_{nk} = p(z = k \mid \mathbf{x}_n, \boldsymbol{\theta}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{k'} \pi_{k'} \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_{k'}, \boldsymbol{\Sigma}_{k'})}$$

M-step:
 $N_k = \sum_n r_{nk}; \hat{\pi}_k = \frac{N_k}{N}; \hat{\boldsymbol{\mu}}_k = \frac{1}{N_k} \sum_n r_{nk} \mathbf{x}_n$
 $\hat{\boldsymbol{\Sigma}}_k = \frac{1}{N_k} \sum_n r_{nk} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)^\top$

Asignación (hard clust.): $\mathbf{z}_n = \arg \max_k r_{nk}$
Frontera GMM: $r_{nk} = r_{nk'},$ lineal si $\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma} \forall k$.

EM & ELBO

Goal: $\max_{\mathbf{w}} \ln p(\mathbf{x}|\mathbf{w}) = \max_{\mathbf{w}} \ln \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}|\mathbf{w})$
Asumiendo $q(\mathbf{z})$:
 $\ln p(\mathbf{x}|\mathbf{w}) = \underbrace{\sum_{\mathbf{z}} q(\mathbf{z}) \ln \frac{p(\mathbf{x}, \mathbf{z}|\mathbf{w})}{q(\mathbf{z})}}_{\mathbb{L}(\mathbf{w}, q)} + \underbrace{\text{KL}(q\|\mathcal{p}(\mathbf{z}|\mathbf{x}, \mathbf{w}))}_{\geq 0}$
 $\therefore \ln p(\mathbf{x}|\mathbf{w}) \geq \mathbb{L}(\mathbf{w}, q)$ (**ELBO lower bound**)

E-step: Maximizar $\mathbb{L}$ respecto q
 $q^* = p(\mathbf{z}|\mathbf{x}, \mathbf{w}^{t-1}) \Rightarrow \text{KL} = 0, \mathbb{L} = \ln p$
M-step: Maximizar $\mathbb{L}$ respecto $\mathbf{w}$
 $\mathbf{w}^t = \arg \max_{\mathbf{w}} \mathbb{E}_{q(\mathbf{z})} [\ln p(\mathbf{x}, \mathbf{z}|\mathbf{w})]$

Convergencia: $\ln p(\mathbf{x}|\mathbf{w}^t) \geq \ln p(\mathbf{x}|\mathbf{w}^{t-1})$
(no-decrece; converge al menos a máx. local)

ELBO expandido: $\mathbb{L} = \mathbb{E}_q [\ln p(\mathbf{x}, \mathbf{z}|\mathbf{w})] + \mathbb{H}(q)$

Demo: $\mathcal{L}$ VAE

$$\ln p(\mathbf{x}|\theta, \phi) = \ln \int p_{\theta}(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \ln \int q_{\phi}(\mathbf{z}|\mathbf{x}) \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})}$$
$$= \ln \mathbb{E}_{q_{\phi}} \left[\frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right]$$

(Jensen) $\geq \mathbb{E}_{q_{\phi}} [\ln \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})}]$
$$= \mathbb{E}_{q_{\phi}} [\ln p_{\theta}(\mathbf{x}|\mathbf{z}) + \ln p(\mathbf{z}) - \ln q_{\phi}(\mathbf{z}|\mathbf{x})]$$
$$= \mathbb{E}_{q_{\phi}} [\ln p_{\theta}(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{q_{\phi}} [\ln \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})}]$$
$$= \mathbb{E}_{q_{\phi}} [\ln p_{\theta}(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

Demo: max varianza equiv min error

$$\text{Var}[z] = \frac{1}{N} \sum_i z_i^2 = \frac{1}{N} \sum_i (\mathbf{x}_i^\top \hat{\mathbf{v}})^2 = \hat{\mathbf{v}}^\top \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}}$$
$$\text{MSE} = \frac{1}{N} \sum_i \|\mathbf{x}_i\|^2 - \hat{\mathbf{v}}^\top \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}}$$
$$= \text{const} - \text{Var}[z]$$
$$\Rightarrow \min \text{MSE} \Leftrightarrow \max \text{Var}[z]. \quad \square$$

Demo: mas K implica log-lik no decrece

Sea $\mathbf{w}_K^*$ óptimo para K clusters. Construir modelo con $K+1$ clusters tomando $\mathbf{w}_{K+1}$ con $\pi_{K+1} = 0$ (degenerar nuevo cluster).
 $\Rightarrow p_{K+1}(\mathbf{X}|\mathbf{w}_{K+1}) = p_K(\mathbf{X}|\mathbf{w}_K^*)$
 $\Rightarrow \ln p(\mathbf{X}|\mathbf{w}_{K+1}^*) \geq \ln p(\mathbf{X}|\mathbf{w}_{K+1}) = \ln p(\mathbf{X}|\mathbf{w}_K^*)$
(el óptimo sobre $K+1$ no puede ser peor que incluir el caso K). $\square$

PCA

Centrar: $\tilde{\mathbf{X}}; \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$
Eigendecomposición: $\hat{\boldsymbol{\Sigma}} \mathbf{u}_l = \lambda_l \mathbf{u}_l;$
 $\mathbf{U}_L = [\mathbf{u}_1, \dots, \mathbf{u}_L] \in \mathbb{R}^{D \times L}$

Encode: $\mathbf{z}_n = \mathbf{U}_L^\top \tilde{\mathbf{x}}_n;$ **Decode:** $\hat{\mathbf{x}}_n = \mathbf{U}_L \mathbf{z}_n + \hat{\boldsymbol{\mu}}$
Error recons.: $\frac{1}{N} \|\tilde{\mathbf{X}} - \tilde{\mathbf{X}} \mathbf{U}_L \mathbf{U}_L^\top\|_F^2 = \sum_{l=L+1}^D \lambda_l$
Var. explicada: $\text{VE}(L) = (\sum_{l=1}^L \lambda_l) / (\sum_{l=1}^D \lambda_l)$

Vía SVD: $\tilde{\mathbf{X}} = \mathbf{U}_X \mathbf{S} \mathbf{V}^\top;$ PC's = cols de $\mathbf{V};$
 $\lambda_l = s_l^2 / N$
Var de z: $\text{Var}[z_l] = \lambda_l$ (coordenadas decorreladas)

Autoencoders

AE determinístico: $\mathbf{z} = f_{\phi}(\mathbf{x}), \hat{\mathbf{x}} = g_{\theta}(\mathbf{z}),$
 $L \ll D$
 $\mathcal{L}_{\text{AE}} = \frac{1}{N} \sum_n \|\mathbf{x}_n - g_{\theta}(f_{\phi}(\mathbf{x}_n))\|^2$
AE lineal $\equiv$ PCA (mismo subespacio); con no-linealidades $\Rightarrow$ reducción no-lineal.

VAE: prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$
Encoder: $q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{\phi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\phi}^2(\mathbf{x})))$
Decoder: $p_{\theta}(\mathbf{x}|\mathbf{z})$

ELBO (a maximizar):
 $\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_{\phi}} [\ln p_{\theta}(\mathbf{x}|\mathbf{z})]}_{\text{reconst.}} - \underbrace{\text{KL}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{regulariz.}}$

KL gaussiano:
 $\text{KL}(\mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))\|\mathcal{N}(\mathbf{0}, \mathbf{I})) = \frac{1}{2} \sum_j (\boldsymbol{\mu}_j^2 + \sigma_j^2 - \ln \sigma_j^2 - 1)$

Reparam.: $\mathbf{z} = \boldsymbol{\mu}_{\phi}(\mathbf{x}) + \boldsymbol{\sigma}_{\phi}(\mathbf{x}) \odot \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
(permite backprop a través del muestreo)

Information Theory

Entropía: $\mathbb{H}(p) = - \sum_i p_i \log p_i \geq 0$
KL divergence (no simétrica, ≥ 0):
 $\text{KL}(q\|p) = \sum_i q_i \ln \frac{q_i}{p_i} = \mathbb{E}_q \left[\ln \frac{q}{p} \right]$
 $\text{KL}(q\|p) = 0 \Leftrightarrow q = p; \quad \text{KL}(q\|p) \neq \text{KL}(p\|q)$

KL gaussianas ($p = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1), q = \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$):
 $\text{KL}(q\|p) = \frac{1}{2} \left[\ln \frac{|\boldsymbol{\Sigma}_1|}{|\boldsymbol{\Sigma}_0|} + \text{tr}(\boldsymbol{\Sigma}_1^{-1} \boldsymbol{\Sigma}_0) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_1^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) \right]$
Caso $p = \mathcal{N}(\mathbf{0}, \mathbf{I})$:
 $\text{KL} = \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - \ln \sigma_j^2 - 1)$

Métricas de Clustering

$\text{WCSS} = \sum_k \sum_{n:r_{nk}=1} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$ ($\downarrow$ mejor)
 $\text{BCSS} = \sum_k N_k \|\boldsymbol{\mu}_k - \bar{\mathbf{x}}\|^2$ ($\uparrow$ mejor)

Silhouette $s(n) \in [-1, 1]; a(n) = \text{dist. intra}, b(n) = \text{dist. al vecino más cercano}$
 $s(n) = \frac{b(n) - a(n)}{\max(a(n), b(n))}; \quad \bar{s} = \frac{1}{N} \sum_n s(n)$

Davies-Bouldin:
 $\text{DB} = \frac{1}{K} \sum_k \max_{k' \neq k} \frac{s_k + s_{k'}}{d(\boldsymbol{\mu}_k, \boldsymbol{\mu}_{k'})}$

Elbow: graficar WCSS vs. K ; codo = K óptimo.
BIC: $-2 \ln \hat{p}(\mathbf{X}) + d_K \ln N$; $\downarrow$ mejor; selecciona K .
Mutual. Info (con labels):
$$I(S_k; Y) = \sum_i \sum_j p_{C_i Y}(i, j) \log \frac{p_{C_i Y}(i, j)}{p_{C_i}(i) p_Y(j)}$$

 $p_{C_i Y}(i, j) = \frac{|c_i \cap y_j|}{N}; \quad I(X; Y) = 0 \Leftrightarrow X \perp Y$
 $I(X; Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X)$
NMI (Normalized Mutual Information):
$$\text{NMI}(S_k, Y) = \frac{2 I(S_k; Y)}{\mathbb{H}(S_k) + \mathbb{H}(Y)} \in [0, 1] \quad (\uparrow \text{ mejor})$$

Demo: EM-GMM - funcion Q y concavidad

E-step — $\mathcal{Q}(\mathbf{w}, \mathbf{w}^{\text{old}})$:
 $p(\mathbf{X}, \mathbf{Z}|\mathbf{w}) = \prod_n \prod_k [\pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]^{z_{nk}}$
 $\ln p(\mathbf{X}, \mathbf{Z}|\mathbf{w}) = \sum_n \sum_k z_{nk} [\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]$
 $\mathcal{Q} = \sum_n \sum_k r_{nk} [\ln \pi_k - \frac{1}{2} \ln |\boldsymbol{\Sigma}_k| - \frac{1}{2} (\mathbf{x}_n - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k)]$

M-step para $\boldsymbol{\mu}_k$: $\nabla_{\boldsymbol{\mu}_k} \mathcal{Q} = 0$
 $\sum_n r_{nk} \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k) = 0 \Rightarrow \hat{\boldsymbol{\mu}}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{N_k}$

M-step para $\boldsymbol{\Sigma}_k$: $\nabla_{\boldsymbol{\Lambda}_k} \mathcal{Q} = 0$ con $\boldsymbol{\Lambda}_k = \boldsymbol{\Sigma}_k^{-1}$:
 $\frac{N_k}{2} \boldsymbol{\Sigma}_k - \frac{1}{2} \sum_n r_{nk} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)^\top = 0$
 $\Rightarrow \hat{\boldsymbol{\Sigma}}_k = \frac{1}{N_k} \sum_n r_{nk} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_k)^\top$

M-step para π_k (Lagrange $\sum \pi_k = 1$):
 $\nabla_{\pi_k} [\mathcal{Q} + \nu (\sum \pi_k - 1)] = 0 \Rightarrow \hat{\pi}_k = \frac{N_k}{N}$

Concavidad de $\mathcal{Q}$: es una suma de $\ln \mathcal{N}$ ponderados; $\ln \mathcal{N}$ es cóncavo en $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \Rightarrow \mathcal{Q}$ cóncava $\Rightarrow$ M-step es único. $\square$

Clustering+++

HAC (Hierarchical Agglomerative):
Bottom-up: empieza con N clusters, fusiona iterativamente los más cercanos $\rightarrow$ dendrograma.
No requiere K a priori.
Linkage: single (min dist), complete (max dist), average, Ward (min Δ WCSS).
DBSCAN: params $\varepsilon, \text{MinPts}$.
Core: $|\mathcal{N}_{\varepsilon}(\mathbf{x})| \geq \text{MinPts}$; *Border:* vecino de core;
Noise: resto.
Clusters = componentes conexas de cores + borders.

for $\mathbf{x}_n$ no visitado **do**
 if $|\mathcal{N}_{\varepsilon}(\mathbf{x}_n)| \geq \text{MinPts}$ **then** nuevo cluster;
 expandir($\mathbf{x}_n$)
 else noise

No requiere K ; detecta formas arbitrarias; sensible a ε .

Spectral Clustering:
Grafo de similitud $\mathbf{W}$; Laplaciano $\mathbf{L} = \mathbf{D} - \mathbf{W}$.
Usar los K autovectores de menor λ de $\mathbf{L}$ como features; aplicar K-Means. Detecta clusters no convexos.

Mixture of Bernoullis ($\mathbf{x} \in \{0, 1\}^D$):
 $p(\mathbf{x}|\mathbf{w}) = \sum_k \pi_k \prod_d \mu_{kd}^{x_d} (1 - \mu_{kd})^{1 - x_d}$
EM igual que GMM; M-step: $\hat{\mu}_{kd} = \frac{\sum_n r_{nk} x_{nd}}{N_k}$

Álgebra Lineal

Traza y ciclo: $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{C}) = \text{tr}(\mathbf{C}\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{C}\mathbf{A})$
 $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top)$ (escalar $\rightarrow$ traza)

Gradientes matriciales clave:
 $\nabla_{\mathbf{A}} \text{tr}(\mathbf{A}\mathbf{B}) = \mathbf{B}^\top$
 $\nabla_{\mathbf{A}} \ln |\mathbf{A}| = \mathbf{A}^{-\top}$ (i.e. $\boldsymbol{\Sigma}^{-1}$ si $\boldsymbol{\Sigma}$ SDP)
 $\nabla_{\mathbf{A}} (\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x}) = -\mathbf{A}^{-\top} \mathbf{x} \mathbf{x}^\top \mathbf{A}^{-\top}$

Identidades para $\boldsymbol{\Sigma}$ (sugeridas en ej. 20):
Si $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$: $\nabla_{\boldsymbol{\Lambda}} \ln |\boldsymbol{\Lambda}| = \boldsymbol{\Lambda}^{-\top} = \boldsymbol{\Sigma}^\top$
 $\nabla_{\boldsymbol{\Lambda}} \text{tr}(\boldsymbol{\Lambda} \mathbf{S}) = \mathbf{S}^\top$

Proyección ortogonal: $\mathbf{P} = \hat{\mathbf{v}} \hat{\mathbf{v}}^\top$ (rango 1, idempotente)
 $\|\mathbf{x} - \mathbf{P}\mathbf{x}\|^2 = \|\mathbf{x}\|^2 - (\mathbf{x}^\top \hat{\mathbf{v}})^2$

Matriz ortogonal $\mathbf{R}$: $\mathbf{R}^\top \mathbf{R} = \mathbf{I}, |\det \mathbf{R}| = 1$
 $\|\mathbf{R}\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{R}^\top \mathbf{R} \mathbf{x} = \|\mathbf{x}\|^2$ (isometría)

Autovalores y traza/det:
 $\text{tr}(\mathbf{A}) = \sum_i \lambda_i; \quad |\mathbf{A}| = \prod_i \lambda_i$
 $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$ incluso si $\mathbf{A}, \mathbf{B}$ no cuadradas

Restricción con Lagrange:
 $\max_{\hat{\mathbf{v}}} \hat{\mathbf{v}}^\top \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}}$ s.t. $\|\hat{\mathbf{v}}\| = 1$
 $\Rightarrow \mathcal{L} = \hat{\mathbf{v}}^\top \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}} - \lambda (\hat{\mathbf{v}}^\top \hat{\mathbf{v}} - 1)$
 $\Rightarrow \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}} = \lambda \hat{\mathbf{v}}$ (autoproblema)
 $\Rightarrow$ máx. varianza = λ_1 con $\hat{\mathbf{v}} = \mathbf{u}_1$

Gaussiana multivariada $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$:
$$\frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

Demo: PCA minimiza error de reconstruccion

Datos centrados ($\hat{\boldsymbol{\mu}} = \mathbf{0}$). $\hat{x}_i = z_i \hat{\mathbf{v}}, z_i = \mathbf{x}_i^\top \hat{\mathbf{v}}$.
 $\text{MSE} = \frac{1}{N} \sum_i \|\mathbf{x}_i - z_i \hat{\mathbf{v}}\|^2$
 $= \frac{1}{N} \sum_i \|\mathbf{x}_i\|^2 - \frac{1}{N} \sum_i (\mathbf{x}_i^\top \hat{\mathbf{v}})^2$
 $= \text{const} - \hat{\mathbf{v}}^\top \left(\frac{1}{N} \mathbf{X}^\top \mathbf{X} \right) \hat{\mathbf{v}}$
 $\Rightarrow$ minimizar MSE $\Rightarrow$ maximizar $\hat{\mathbf{v}}^\top \hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}}$.
Con restricción $\|\hat{\mathbf{v}}\| = 1$ (Lagrange):
 $\hat{\boldsymbol{\Sigma}} \hat{\mathbf{v}} = \lambda \hat{\mathbf{v}}$; la varianza = λ .
 $\Rightarrow \hat{\mathbf{v}}^* = \mathbf{u}_1$ (autovector de $\lambda_{\max}$).

Demo: PCA invariante a roto-traslaciones

$\mathbf{x}'_i = \mathbf{R} \mathbf{x}_i + \mathbf{b}$
Traslación: $\mathbf{x}'_i = \mathbf{x}_i + \mathbf{b} \Rightarrow \boldsymbol{\mu}' = \boldsymbol{\mu} + \mathbf{b}$.
 $\tilde{\mathbf{x}} = \mathbf{x}'_i - \boldsymbol{\mu}' = \mathbf{x}_i + \mathbf{b} - (\boldsymbol{\mu} + \mathbf{b}) = \mathbf{x}_i - \boldsymbol{\mu}$.
Rotación: $\tilde{\mathbf{x}}'_i = \mathbf{R} \tilde{\mathbf{x}}_i, \mathbf{R}$ de rotación ($\perp, \det = 1$).
 $\hat{\boldsymbol{\Sigma}}' = \mathbf{R} \hat{\boldsymbol{\Sigma}} \mathbf{R}^\top = \mathbf{R} \hat{\boldsymbol{\Sigma}} \mathbf{R}^\top$
Si $\hat{\boldsymbol{\Sigma}} \mathbf{u}_l = \lambda_l \mathbf{u}_l \Rightarrow \hat{\boldsymbol{\Sigma}}'(\mathbf{R} \mathbf{u}_l) = \lambda_l (\mathbf{R} \mathbf{u}_l)$
Coords reducidas:
 $z'_l = (\mathbf{R} \mathbf{u}_l)^\top (\mathbf{R} \tilde{\mathbf{x}}_i) = \mathbf{u}_l^\top \mathbf{R}^\top \mathbf{R} \tilde{\mathbf{x}}_i = z_l$.